Cart2Insights: Decoding E-Commerce Performance

Olist Brazilian e-commerce, January 2017 to August 2018

GUVI x HCL Capstone | github.com/sathish-p-0406/cart2insights

Submitted By – Sathish P

R$15.69M
revenue

97,910
orders

4.12 / 5
average review

From 9 raw CSVs to a decision-ready database

1

Clean

Removed duplicates and orphan rows, filled missing categories

2

Query

SQLAlchemy on SQLite: KPIs, CTEs, subqueries, window functions

3

Engineer

Order, customer, seller and item feature tables

4

Explore

Six EDA chart groups and three hypothesis tests

5

Deliver

Interactive Streamlit dashboard with six sections

9 base tables with primary and foreign keys

4 feature tables for analysis

20 months of orders analysed

R$15.69M in revenue from 97,910 orders

R$15.69M
Total revenue

97,910
Orders

94,707
Customers

3,029
Sellers

R$160.22
Average order value

4.12
Average review score (out of 5)

January 2017 to August 2018, excluding canceled and unavailable orders

SP, RJ and MG lead sales; November 2017 was the peak

TOP STATE BY REVENUE

~R$5.77M

São Paulo (SP), 40,500 delivered orders

PEAK MONTH

Nov 2017

Black Friday drove the highest monthly revenue

OBSERVATION

São Paulo leads with ~R$5.77M from 40,500 delivered orders, ahead of RJ and MG. Monthly revenue peaked in November 2017.

INTERPRETATION

Demand is led by the Southeast and spikes around Black Friday.

BUSINESS IMPACT

Prioritise stock and delivery capacity in these states, and plan inventory and campaigns ahead of November.

Only 3.1% of customers come back

OBSERVATION

93,099 customers bought once; only 2,997 (about 3.1%) ordered again.

INTERPRETATION

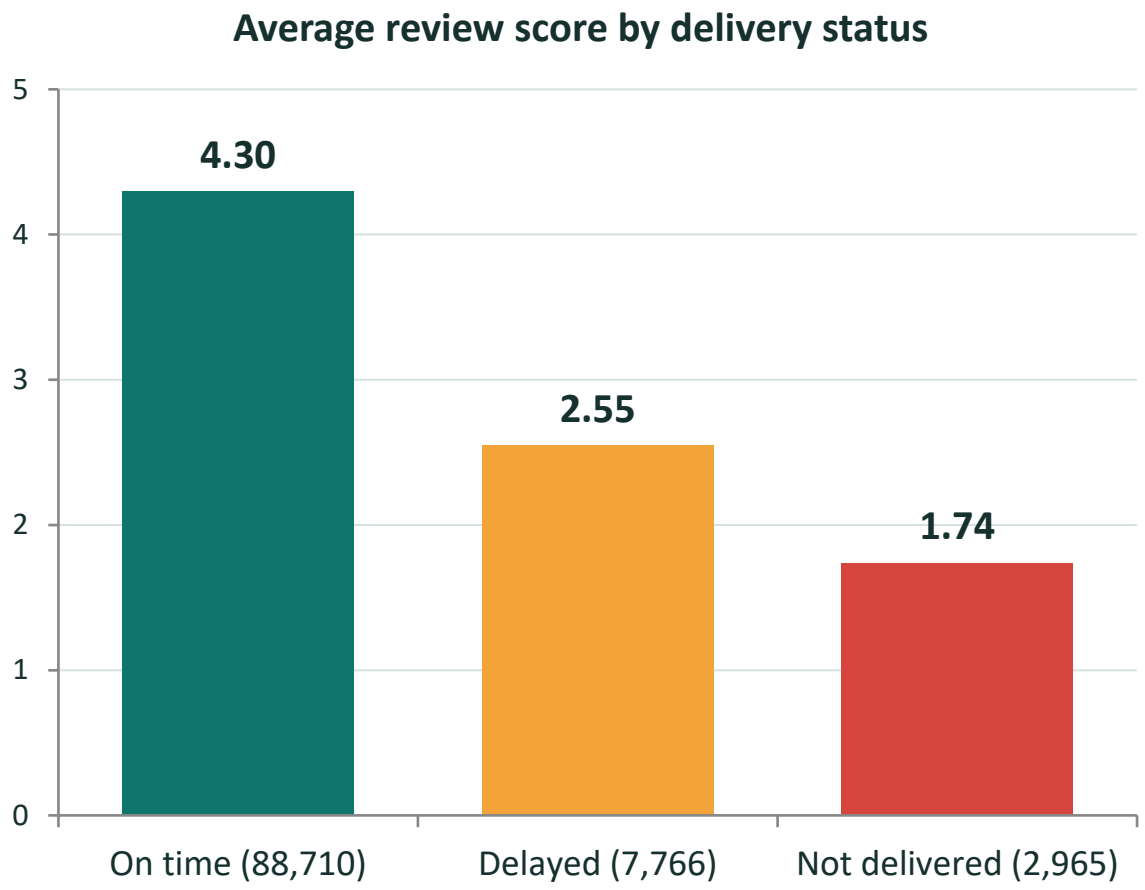
Acquisition works, but loyalty is low: most buyers treat the marketplace as a one-off purchase.

BUSINESS IMPACT

A small lift in repeat rate adds revenue with no new acquisition cost. Target one-time buyers with retention campaigns.



Late delivery drops reviews from 4.30 to 2.55



OBSERVATION

88,710 orders arrived on time, 7,766 late and 2,965 were never delivered. Average review: 4.30, 2.55 and 1.74.

INTERPRETATION

Delivery time matters more than price: review score correlates −0.33 with delivery days but only −0.04 with order value.

BUSINESS IMPACT

Late and missing deliveries are the biggest satisfaction risk. Fix slow lanes and set realistic delivery estimates.

Statistics confirm delivery delay has the largest effect

Welch t-test

Do delayed orders get lower review scores than on-time ones?

$d = 1.22$

Cohen's d; $t = -89.77$, $p < 0.001$

Large effect: delayed orders are rated far lower

One-way ANOVA

Does order value differ across 51 product categories?

$\eta^2 = 0.098$

$F = 208.07$, $p < 0.001$

Category explains ~10% of order-value variation

Chi-square test

Is payment method linked to order status?

$V = 0.028$

Cramér's V; $\chi^2 = 154.48$, $df = 6$, $p < 0.001$

Significant, but the effect is negligible

Robustness: Mann-Whitney and Kruskal-Wallis tests agree with the parametric results. Limitation: Levene's test shows unequal variances across categories.

Order value ranges from R$81 to R$1,287 by category

OBSERVATION

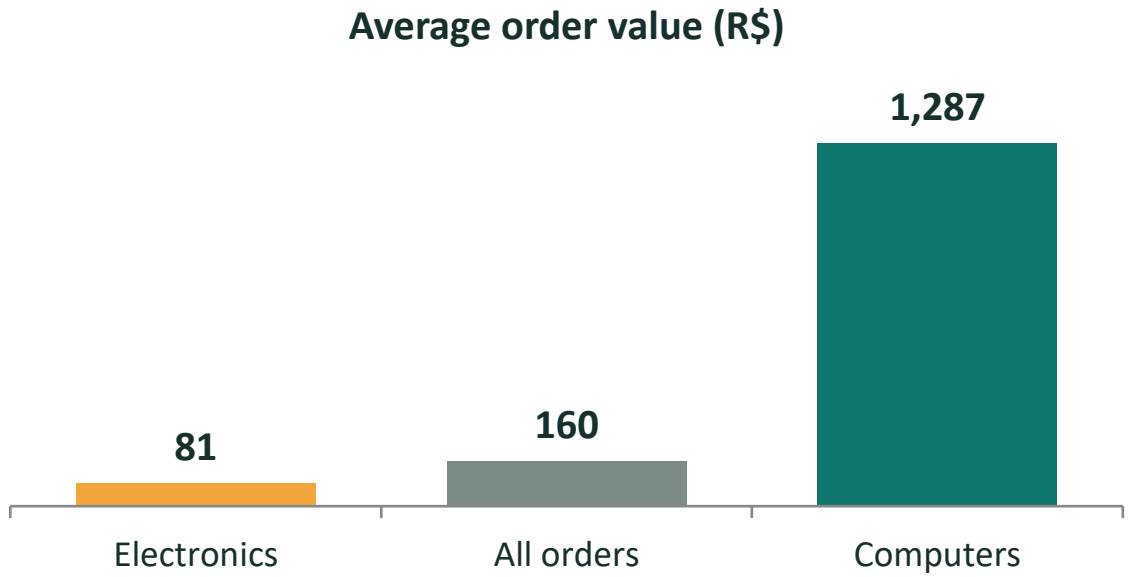
Average order value spans R$81 (electronics) to R$1,287 (computers). Voucher orders fail 3.62% of the time vs ~1.2% for other payment types.

INTERPRETATION

Category drives basket size (about 10% of variation). The voucher gap is statistically real but small overall (V = 0.028).

BUSINESS IMPACT

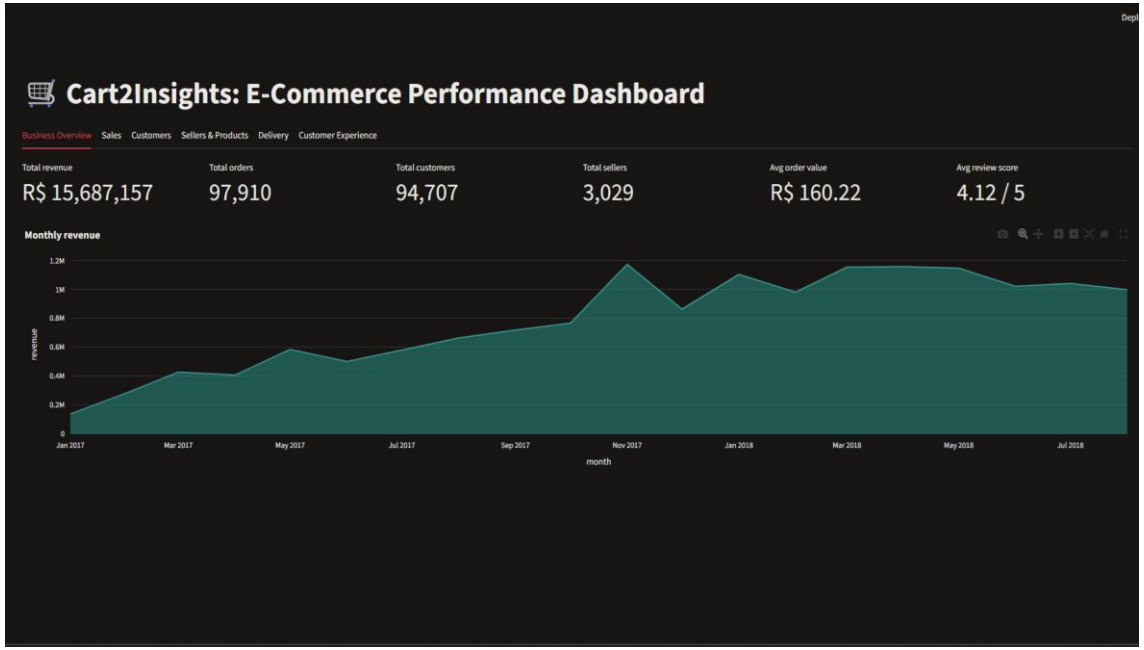
Promote high-value categories and bundle low-value ones. Audit the voucher checkout and fulfilment flow.

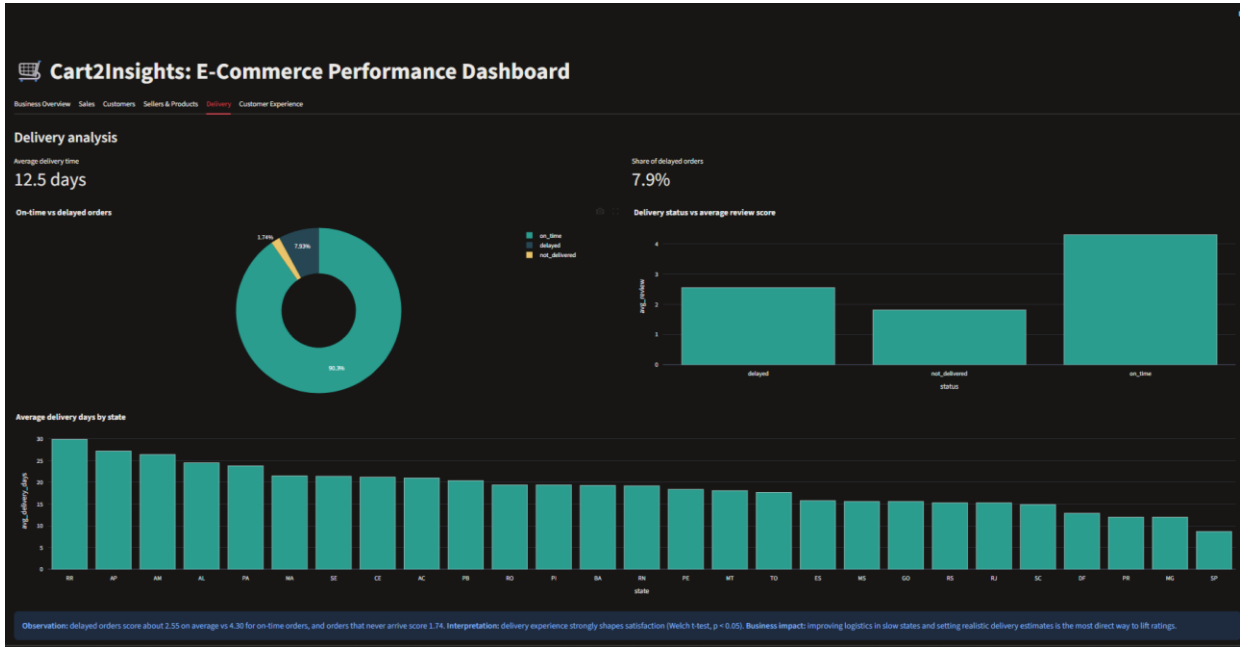


3.62%
voucher orders canceled or unavailable

~1.2%
other payment types

A six-section interactive Streamlit dashboard





Business Overview | **Sales** | **Customers** | **Sellers & Products** | **Delivery** | **Customer Experience**

Built with Streamlit and SQLAlchemy; code and README are on GitHub

Four actions to grow revenue and satisfaction

1

Improve logistics in slow states and set realistic delivery estimates

Delayed orders average 2.55 stars vs 4.30 on time

2

Run retention campaigns for the ~97% one-time buyers

Only 3.1% of customers (2,997) ordered again

3

Promote high-value categories and bundle in low-value ones

Average order value: computers R$1,287 vs electronics R$81

4

Investigate the voucher checkout and fulfilment flow

Voucher orders: 3.62% canceled or unavailable vs ~1.2% for other payments

Code, notebooks and README: github.com/sathish-p-0406/cart2insights